

Marketing was built on the page view. The funnel now pays for the **consented login**.

Consent Mode v2, the decay of third-party cookies, and first-party identity moved the funnel's unit of value from the anonymous page view to the consented, logged-in identity. A page view **depreciates**: cookie-fragile, anonymous, disposable. A consented login **appreciates**: durable, addressable, worth money — and the only thing an AI agent or an AI answer engine can act on. Most marketing apps still pivot on the page view. We build on the login.

For: Marketing ops, performance media, and analytics engineering

Status: Built (GA4 push + audiences) · Direct Google push routes via the Data Manager API — plan: [SEGMENTS-GOOGLE-CONNECT.md](#)

Date: 2026-07-07 Depends on: [FEATURE-SPEC-UA-MIGRATION.md](#) · [ARCHITECTURE.md](#)

0. TL;DR

Consent Mode v2, the decay of third-party cookies, and first-party identity moved the funnel's unit of value from the **anonymous page view** to the **consented, logged-in identity**. A page view depreciates — cookie-fragile, anonymous, disposable. A consented login appreciates — durable, addressable, worth money, and the only thing an AI agent or an AI answer engine can act on.

This spec defines the pipe that turns a **Shopify customer segment** into a **Google Smart Bidding signal** without a CSV, without an unconsented identifier, and without a static conversion value. A revenue-weighted, consent-gated user property flows from Xano → the Cloudflare Worker → GA4 (Measurement

Protocol) → a GA4 audience → Google Ads, where Smart Bidding optimizes to the value of the *identity*, not the *click*.

One-line recommendation: *Stop bidding on page views. Bid on consented, revenue-weighted segments — computed server-side, gated on consent, and delivered to Google as a first-party signal.*

1. Why the funnel got repriced

OLD UNIT OF VALUE (PAGE VIEW)	NEW UNIT OF VALUE (CONSENTED LOGIN)
Anonymous — no durable identity to bid on	First-party identity, keyed by a portable UUID
Cookie-fragile — cleared, blocked, or expired	Server-side record, survives the browser
Static conversion value set once	Value on every event, recomputed per order
Invisible to consent — "consent laundering" risk	Consent is a first-class field, checked before every push
Opaque to agents and answer engines	Machine-addressable — an agent can act on it

Three forces made the page view a wasting asset: **Consent Mode v2** (four required signals in the EEA — `ad_storage` , `analytics_storage` , `ad_user_data` , `ad_personalization`), the collapse of third-party cookies, and the shift of Google's own bidding toward first-party, value-based signals. Most marketing apps still pivot on the page view. This one builds on the login.

2. The pipe — Shopify Segment → Smart Bidding

SHOPIFY		XANO		CLOUDFLARE WORKER		GOOGLE
customer tags		source of truth		projection + consent gate		GA4 + Ads
+ Segments	→	crm tags (segment,	→	on tag change:	→	Measurement
+ metafields		tier, campaign,		build user properties,		Protocol → GA
+ Customer		consent) + revenue		check consent_marketing,		user propert
Events (JIT)		weight per identity		push server-side		GA4 audience
						Ads Smart Bi

The invariant: the segment is *computed in your data layer*, not in the ad platform. Google receives a consent-qualified, revenue-weighted signal keyed to a first-party identity — never a raw customer list, never an unconsented identifier.

2.1 What each layer owns

LAYER	OWNS	EMITS
Shopify	Customer tags, Segments, <code>custom.crm_segment</code> metafield, Customer Events (JIT triggers)	Order value, segment membership, purchase events
Xano	The system of record: CRM tags by category (<code>status</code> , <code>tier</code> , <code>segment</code> , <code>campaign</code> , <code>consent</code> , <code>marketing</code>), revenue weight per identity, consent state	Canonical, versioned user record keyed by UUID
Cloudflare Worker	Projection + the consent gate . On every tag change it builds GA4 user properties and refuses to push if <code>consent_marketing ≠ granted</code>	GA4 Measurement Protocol calls, fail-closed
GA4	User-scoped custom dimensions → Audiences	Audiences shared to Google Ads
Google Ads	Smart Bidding (tROAS / tCPA) consuming the audience + value	Bids weighted to the identity's value

3. The signals that cross the wire

The worker pushes these GA4 **user properties** on every tag change (from the dashboard, the 15-minute Shopify sync, or the admin API):

USER PROPERTY	SOURCE (CRM TAG CATEGORY)	EXAMPLE	ROLE IN BIDDING
crm_segment	segment	high_value, returning	Audience membership
crm_tier	tier	vip	Value tier / bid multiplier intent
crm_campaign	campaign	summer_2026	Campaign audience scoping
crm_status	status	active	Suppression / eligibility
consent_marketing	consent	granted denied	Gate — no grant, no push
crm_revenue_band	derived (order history)	p90	Revenue weighting for tROAS

Events: `crm_tags_updated` (dashboard tag change; carries `tags_added` , `tags_removed` , `campaign_tags`) and `crm_sync` (`source: shopify_cron`). Both carry only consented, server-verified state.

4. Consent is the gate, revenue is the weight

Two rules make this legal *and* effective:

1. **Consent gates the push (fail-closed).** Before any Measurement Protocol call, the worker checks `consent_marketing` . Denied or unknown → the signal is not

sent. Consent is stored server-side in Xano, not read from a browser cookie, so an outage or a cleared cookie cannot silently reopen the pipe. When a customer opts out, every connected destination is notified in the same request — no stale audience keeps bidding on someone who left.

2. **Revenue is the weight, not the headcount.** A segment isn't "how many logins" — it's "how much value." The worker attaches a revenue band per identity so Google's **value-based Smart Bidding (tROAS)** optimizes toward the identities that actually pay, instead of treating every consented login as equal. This is the difference between a remarketing list and a *bidding signal*.

Compliance here is not a cost center. A consent-gated, revenue-weighted signal is *more* valuable to Smart Bidding than an unconsented bulk list — because it is legal to use, durable, and priced to value. Enforcement is the reason the signal is safe to send.

5. Just-in-time membership (Shopify Customer Events)

Batch segment sync is the floor, not the ceiling. Shopify **Customer Events** (Web Pixel / server pixel) let the segment update *at the moment of behavior*.

- A `checkout_completed` event raises the identity's revenue band → the next GA4 push moves them up a bidding tier.
- A high-intent browse pattern adds a `campaign` tag → JIT audience entry, no waiting for the 15-minute cron.
- An opt-out event flips `consent_marketing` → the worker suppresses the next push immediately.

JIT keeps the bidding signal fresh to the event, while the cron sweep guarantees eventual consistency and heals any drift.

6. Setup (operator steps)

1. **GA4 Measurement Protocol** — set `GA4_MEASUREMENT_ID` (`G-XXXXXXXXXX`) and the API secret in the app config.
 2. **User-scoped custom dimensions** in GA4 for `crm_segment` , `crm_tier` , `crm_campaign` , `consent_marketing` , `crm_revenue_band` .
 3. **GA4 Audiences** — e.g. *VIP*: `crm_tier` contains "vip" ; *At-Risk*: `crm_segment` contains "at_risk" ; *Marketing Opted-In*: `consent_marketing` equals "granted" . Audiences auto-share to Google Ads.
 4. **Google Ads** — turn on value-based Smart Bidding (tROAS) on the campaigns consuming the shared audiences.
 5. **Shopify metafield mirror** — confirm `customer.metafield.custom.crm_segment` is populated so the same segment is queryable inside Shopify Segments and Flow.
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7. Status & what's next

CAPABILITY	STATE
CRM tags → GA4 user properties (server-side)	Live
Consent gate (fail-closed on <code>consent_marketing</code>)	Live
GA4 audiences → Google Ads sharing	Live (native GA4 → Ads link)
Revenue-weighted band (<code>crm_revenue_band</code>)	Live
JIT membership via Shopify Customer Events	Live
Direct push to Google Ads Customer Match / offline conversions	Planned — via the Data Manager API (plan)

The direct path (Customer Match audiences + value-based conversion ingestion, hashed first-party identifiers under consent) is the next increment — and it no longer runs through the Google Ads API. Google disabled Ads API Customer Match uploads on April 1, 2026; the successor channel is the **Data Manager API**, planned in `SEGMENTS-GOOGLE-CONNECT.md` . Until that ships, GA4's native audience share carries the signal.

8. Theme the surfaces this bids on

Every surface a bid lands on — the storefront, the login, the account, the dashboard — is themed from one place. Open the store this stack is built on: → www.crm-sync.dev — the live store this stack is built on.

One primary color and token set key Shopify, Webflow, and every embed, so the consented login you bid on looks like one brand across the whole funnel. The design system behind those surfaces is documented in `UI-COMPONENT-REGISTRY.md` .

The reframe: stop paying for the anonymous page view. Compute the segment in your data layer, gate it on consent, weight it by revenue, and hand Google a first-party signal it can actually bid on. That is the funnel, repriced.
